

Nikola Vuk Maruszewski

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🇺🇸 United States Citizen

EDUCATION

Georgia Institute of Technology

PhD, Computer Science

Atlanta, GA

Aug 2025 – Expected May 2029

- **GPA:** 4.00/4.00
- **Advisor:** Josiah Hester

Northwestern University

Master of Science, Computer Engineering

Evanston, IL

April 2024 – Jun 2025

- **GPA:** 4.00/4.00
- **Thesis:** Improved Prefetching Techniques for Linked Data Structures
- **Committee:** Nikos Hardavellas (advisor), Peter Dinda, Russ Joseph

Northwestern University

Bachelor of Science, Computer Science

Evanston, IL

Sep 2022 – Jun 2025

- **GPA:** 4.00/4.00, *summa cum laude*
- Dean's list with High Honors, all quarters

PUBLICATIONS

Magnetic Tunnel Junctions for Timekeeping in Intermittent Computing Systems

Preprint, Jul 2026

Nikola Vuk Maruszewski, Jordan Athas, Allison Fleming, Christian Duffee, Eren Yildiz, Saad Ahmed, Yaman Sangar, Pedram Khalili, and Josiah Hester. *arXiv Hardware Architecture (cs.AR)* [arXiv:2607.23000](https://arxiv.org/abs/2607.23000), July 2026.

SEQC: Stratify-Elaborate Quantum Compilation for Modular Architectures

Sep 2026

Mingyoung Jessica Jeng*, Nikola Vuk Maruszewski*, Mu-Te Lau, Connor Selna, Michael Gavrincea, Kaitlin N. Smith, and Nikos Hardavellas. In *Proceedings of the IEEE International Conference on Quantum Computing and Engineering (QCE)*, Toronto, ON, Canada, September 2026.

(* denotes equal contribution)

Media Coverage:

- The Quantum Insider. [Researchers Say Quantum Compiler Boosts Speed And Reliability For Chiplet-Based Modular Systems](#). January 22, 2025
- Semiconductor Engineering. [Parallelized Compilation Pipeline Optimized for Chiplet-Based Quantum Computers](#). January 21, 2025

Improved Prefetching Techniques for Linked Data Structures

M.S. Thesis, Jun 2025

Nikola Vuk Maruszewski. M.S. Thesis, Northwestern University, Technical Report NU-CS-2025-05, Evanston, IL, June 2025. DOI: <https://doi.org/10.21985/n2-bsav-a158>. Also, *arXiv Hardware Architecture (cs.AR)* [arXiv:2505.21669](https://arxiv.org/abs/2505.21669), June 2025.

EXPERIENCE

Graduate Researcher

Georgia Institute of Technology

Aug 2025 – Present

Atlanta, GA

Researcher in the [Ka Moamo Lab](#).

- Developed FLINT, a batteryless timekeeping system for intermittent computing using magnetic tunnel junction (MTJ) arrays; currently under review.
- Awarded a [CRNCH fellowship](#) to support my MTJ work.
- Developed a reactive Edge-AI system for multimodal, on-device inference across heterogeneous compute (CPU/GPU/HTP), including deployment of Whisper, LLM, and VLM models on Qualcomm QCS6490 platforms.

Graduate Teaching Assistant

Georgia Institute of Technology

Aug 2026 – Present

Atlanta, GA

Teaching assistant for CS 6210: Advanced Operating Systems.

- Held office hours and answered student questions on course material and projects.
- Helped develop and grade assignments and exams.

Machine Learning Developer

Caterpillar, Inc.

Sep 2024 – Present

Remote (consulting)

Worked part-time in the Autonomy and Automation Division on Machine Learning pipelines.

- Brought project from prototype to MVP and led initial production deployment.
- Partnered with lead architect on major system design decisions.
- Designed and implemented key pipeline features end-to-end.
- Onboarded and mentored new team members.

Undergraduate Researcher

PARAG@N Lab

Sep 2022 – Jun 2025

Evanston, IL

Led a research project to design improved Quantum Systems software.

- Designed and programmed a quantum compiler to optimize quantum circuits for emerging quantum computer topologies.
- Created a development framework and tools for further quantum systems research.
- Student leader of the project while an undergraduate student.

Software Engineering Intern

Caterpillar, Inc.

Jun 2024 – Aug 2024

Peoria, IL

Worked in the Autonomy and Automation Division on computer vision and data processing.

- Designed and implemented a new data warehouse and processing pipeline in Python.
- Built distributed concurrency control for distributed compute using ZooKeeper.
- Developed a smartphone vehicle calibration system using OpenCV in Python.

Teaching Assistant

Northwestern University

Jun 2023 – Jun 2024

Evanston, IL

Acted as an undergraduate peer mentor for CS 321: Programming Languages and CS 213: Intro to Computer Systems.

- Held several office hours each week.
- Answered questions, both synchronously in office hours and asynchronously on a Piazza message board.
- Helped develop and test homeworks and exams.

Campus Ambassador

Ansys, Inc.

Sep 2023 – Jun 2024

Evanston, IL

Acted as the Campus Ambassador for Ansys at Northwestern.

- Researched and scheduled outreach meetings with campus groups to promote Ansys tools.
- Organized and ran lunch info sessions, handling logistics and coordination.

AWARDS AND HONORS

CRNCH Fellowship | *Georgia Institute of Technology*

Aug 2026

Fellowship awarded by Georgia Tech's Center for Research into Novel Computing Hierarchies (CRNCH).

Outstanding CS Senior | *Northwestern University*

May 2025

Given to the top members of the graduating Computer Science class at Northwestern.

Tau Beta Pi (TBPI) | *Northwestern University*

Nov 2024

Engineering honor society. IL-Gamma chapter.

McCormick Summer Research Award | *Northwestern University*

May 2023

Title: "A Compiler for Quantum Chiplets." Advised by Nikos Hardavellas.

Northwestern Academic Year Undergraduate Research Award | *Northwestern University*

Feb 2023

Title: "A Compiler for Quantum Chiplets." Advised by Nikos Hardavellas.

Dean's List with High Honors | *Northwestern University*

Dec 2022 — Jun 2025

Awarded each quarter to students with a 4.00 GPA. Received every quarter at Northwestern.

RESEARCH GRANTS

- McCormick Summer Research Award** | *Northwestern University* May 2023
Title: “A Compiler for Quantum Chiplets.” Advised by Nikos Hardavellas. \$4500 (supplemented to \$8000).
- Northwestern Academic Year Undergraduate Research Award** | *Northwestern University* Feb 2023
Title: “A Compiler for Quantum Chiplets.” Advised by Nikos Hardavellas. \$1000.

TALKS AND PRESENTATIONS

- A Compilation Framework for Chiplet-Based Quantum Computing Systems** Sep 2023
Given at Northwestern University.
- Quantum Computing Research at PARAG@N** May 2023
Lecture given for a class session of COMP_ENG 456 at Northwestern University.